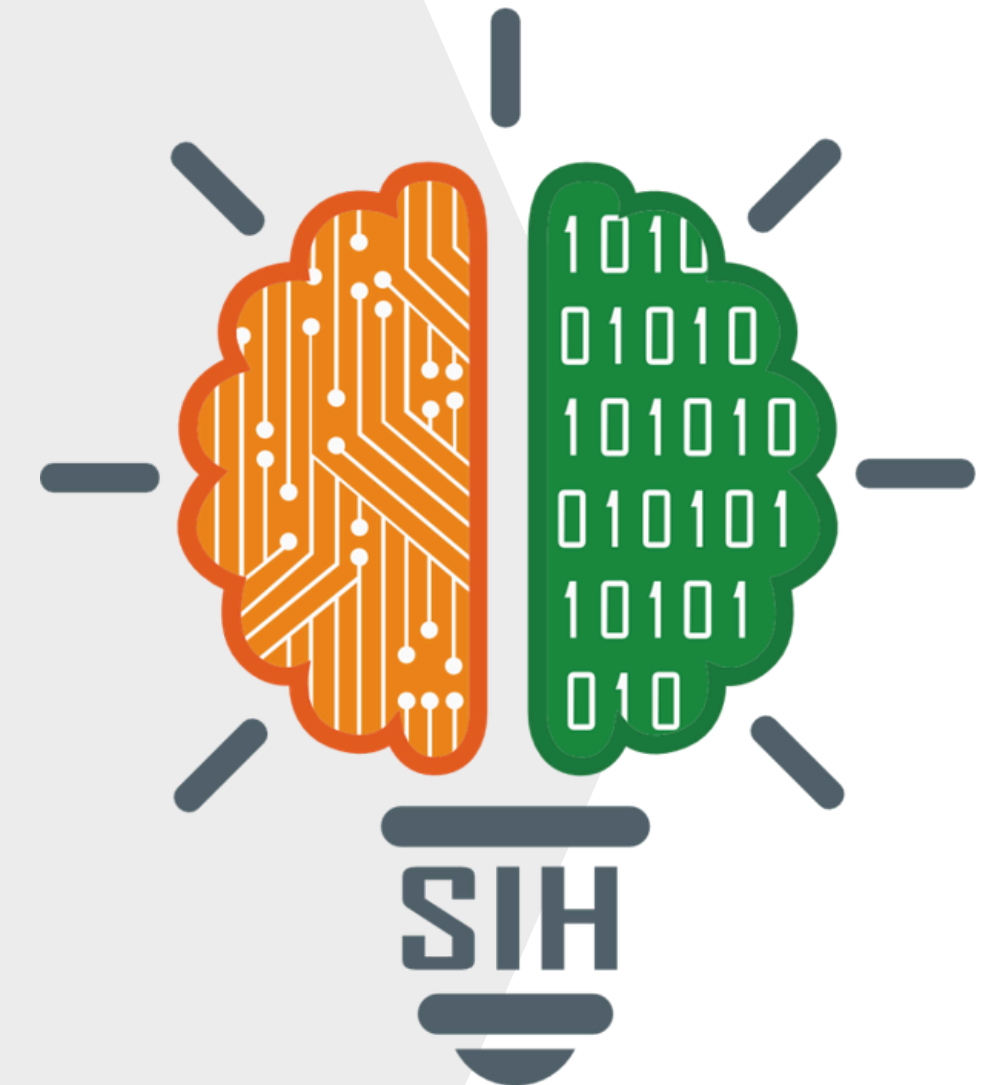


TITLE PAGE

- **Problem Statement ID –** 26189
- **Problem Statement Title-** AI-Powered Criminal Network Analysis System
- **Theme-** Blockchain & Cybersecurity
- **PS Category-** Software
- **Team ID-**
- **Team Name -** Genesis

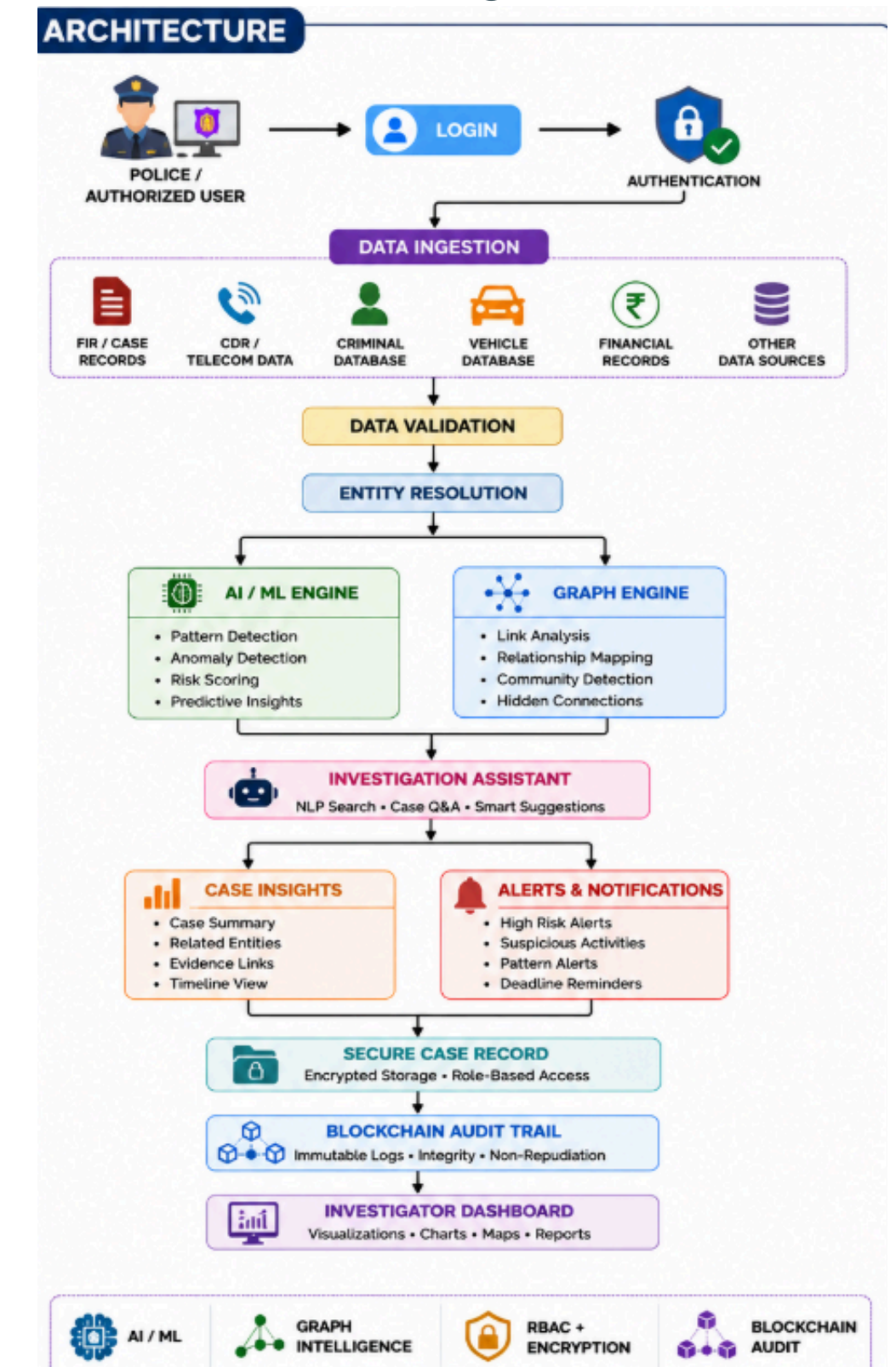


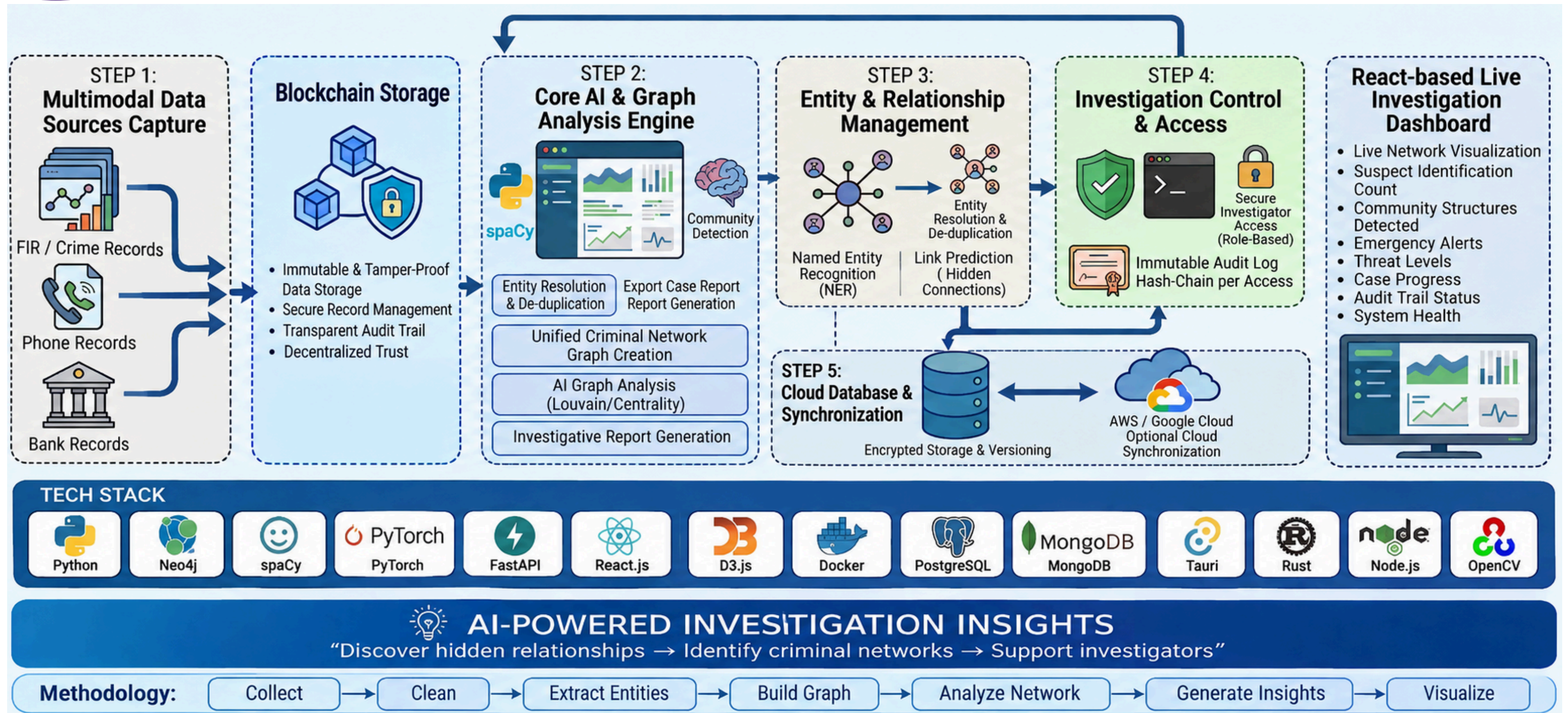
Problem Existing

- **Fragmented Crime Data** : FIRs, CDRs, criminal records, financial data & vehicle details are stored across different systems.
- **Manual Investigation** : Officers spend hours searching, comparing and connecting information from multiple sources.
- **Hidden Connections** : Relationships between suspects, locations, phone numbers, vehicles & cases are difficult to identify manually.
- **Data Security Risks** : Sensitive investigation data requires strong access control, encryption and tamper-proof records.

Proposed Solution

- **AI-Powered Crime Intelligence** : Uses AI/ML to analyze large volumes of investigation data and automatically identify patterns.
- **Unified Data Integration** : Connects FIR, CDR, criminal, vehicle, financial and other authorized datasets in one platform.
- **Intelligent Link Analysis** : Builds relationship graphs between persons, cases, phone numbers, locations, vehicles and organizations.
- **Natural Language Investigation** : Officers can ask questions in simple language and receive relevant case insights from authorized data.
- **Secure & Tamper-Proof Evidence** : Encryption, role-based access and blockchain-based audit trails protect sensitive investigation records.





FEASIBILITY :

Technical :

- **Proven Technology Stack** – Python/FastAPI, PostgreSQL, Neo4j, NLP/LLMs and Graph Analytics provide a scalable and extensible architecture.
- **Multi-Source Integration** – Supports FIRs, CDRs, financial transactions, surveillance reports and criminal databases.
- **Secure & Explainable AI** – Role-based access, encryption, audit logs and confidence scores support responsible investigation.

Innovation :

- **AI-Powered Network Discovery** – Automatically extracts entities and relationships and converts fragmented investigation data into an interactive knowledge graph.
- **Cross-Case Intelligence** – Identifies hidden links, key intermediaries and connections across seemingly unrelated cases.

Operational :

- **Investigator-Friendly** – Search, network visualization, timeline analysis and automated pattern detection reduce manual data correlation.

Economic :

- **Software-based solution** with minimal additional hardware requirements and reduced investigation effort.

VIABILITY :

Market :

Growing demand for AI-driven intelligence and data analytics to correlate large volumes of law-enforcement data and uncover hidden relationships.

Policy & Revenue :

Supports Digital India, smart policing and data-driven governance; suitable for deployment through government and institutional procurement.

Investment & ROI :

Primarily requires software development and secure infrastructure investment while reducing manual investigation and data-correlation effort.

Scalability & Retention :

Can scale from individual case analysis to district, state and multi-agency investigations as new data sources are integrated.



TECHNICAL CHALLENGES :

Risk : Investigation data may be incomplete, inconsistent or unstructured, affecting entity and relationship extraction.

Mitigation : NLP preprocessing, entity resolution, confidence scoring and human-in-the-loop verification ensure reliable results.



BUSINESS / USER CHALLENGES :

Risk : Investigators may hesitate to trust AI-generated connections when handling sensitive criminal data.

Mitigation : Explainable AI, evidence traceability, role-based access and audit logs ensure transparency. Final decisions remain with authorized investigators.



POTENTIAL IMPACT ON THE TARGET AUDIENCE



1. Increased Risk & Threat

Hidden criminal networks lead to more organized crimes, causing fear, insecurity and loss of trust in law enforcement.



2. Delayed Justice

Slow investigations and fragmented data result in delayed arrests and prolonged victim suffering.



3. Higher Financial & Emotional Burden

Victims and families face financial loss, mental stress, and long legal processes.



4. Spread of Crime in Society

Unidentified networks continue operating, leading to more crimes and unsafe communities.



5. Reduced Quality of Life

Constant fear, property damage, and community instability affect overall quality of life.



6. Loss of Public Trust

Ineffective crime control reduces confidence in the justice system and governance.



BENEFITS OF THE SOLUTION



1. Safer Communities

Identifying and disrupting criminal networks reduces crime and makes communities safer.



2. Faster & Effective Investigations

Automated analysis and connection discovery speed up investigations and improve accuracy.



3. Economic Savings

Preventing crimes reduces losses, legal costs, property damage and law enforcement expenses.



4. Better Resource Utilization

Helps agencies focus on high-impact targets, optimizing time, manpower and resources.



5. Strengthened Justice System

Data-driven insights lead to better decisions, higher conviction rates and public trust.



6. Supports Sustainable & Peaceful Society

Early detection and prevention reduce crime, promote harmony and support long-term social well-being.



From identifying individual criminals to **understanding and disrupting the entire criminal network.**

Research Area	What Existing Research Shows
1. Social & Graph Network Analysis	Represents criminals, organizations, locations, communications, and transactions as interconnected nodes and edges to uncover hidden relationships and network structures.
2. AI / Machine Learning	Uses ML and graph-based AI to detect patterns, identify hidden links, predict associations, and prioritize suspicious network activity.
3. Community & Key Player Detection	Network algorithms identify criminal groups, influential individuals, coordinators, and critical nodes within an organization.
4. Blockchain & Secure Data Provenance	Blockchain can provide tamper-evident records, traceable transactions, and trustworthy data sharing between authorized investigative agencies.
5. Cybersecurity & Privacy	Protects sensitive criminal intelligence through encryption, access control, authentication, secure data sharing, and privacy-aware analysis.

References

Patent ID	Patent / Topic	Relevance to Criminal Network Analysis
US20120084288A1	Criminal Relationship Analysis and Visualization	Connects suspects, crimes, phone records, locations, events, etc., and visually represents their relationships.
US11651461B1	Artificial Intelligence Crime Linking Network	Uses AI for criminal network and crime-link analysis.
US20140195984A1	Analytic Frameworks for Persons of Interest	Uses network graphs and law-enforcement, financial, and public data to identify suspicious persons and communities.
CN104408149A	Criminal Suspect Mining Association Method and System Based on Social Network Analysis	Uses social-network graphs and community discovery to identify potentially associated criminals.
CN114915468A	Intelligent Analysis and Detection Method for Network Crime Based on Knowledge Graph	Uses knowledge graphs for intelligent crime analysis and detection.